Lucas C.M. Ulibarri

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EDUCATION

BAYLOR UNIVERSITY- Waco, TX **Dec 2027**
Bachelor of Science in Computer Science (Software Engineering)

- **Relevant Coursework:** Linear Algebra, Calculus I & II, Programming I & II, Data Structures, Algorithms, Software I

Technical Skills

- **Languages:** C++, C, Java, Python, Bash, Swift
- **Graphics/Frameworks:** Vulkan, GLSL, SwiftUI, AppKit, CoreGraphics
- **Tools:** Git, GitHub, CMake, Linux, Visual Studio, Xcode, VIM/NVIM
- **Concepts:** Data Structures & Algorithms, object-Oriented Programming, Memory Management

EXPERIENCE

BAYLOR UNIVERSITY, ITS NETWORKING - Waco, TX **Fall 2024-Present**
Student Worker

- Fielded hundreds of network activation help tickets across campus, reducing employee workload and allowing staff to focus on higher-priority technical tasks.
- Mapped WiFi dead zones through comprehensive wireless network surveys, contributing to 70% campus-wide improvements in connectivity.
- Provisioned numerous network jacks in campus buildings, systematically labeling cables to enhance wired network management and troubleshooting efficiency.
- Rewired network closets by deactivating and rerouting patch cables, improving organization and accessibility for future maintenance.

BAYLOR UNIVERSITY, GOLDEN WAVE BAND - Waco, TX **Present**
Squad Leader, Saxophone Section

- Helm the Golden Wave Band's Saxophone Section as Squad Leader, providing leadership and support for fellow student musicians.
- Convey section expectations, rehearsal information, and performance-related updates to help members arrive prepared and coordinated.
- Nurture a collaborative, welcoming section culture by helping teammates navigate rehearsals, performances, and campus-band involvement.

Personal Projects

Rabbit Hole | Wacode Hackathon - Team Project **Oct 2026**
TypeScript, Next.js, React Flow, Python, Flask, SQLite

- Co-developed a full-stack learning platform connecting diagnostic assessments, interactive concept roadmaps, lessons, and quizzes through Flask REST APIs.
- Collaborated on prerequisite-based learning workflows with React Flow visualizations, server-side grading, and concept unlocking as learners complete lessons.
- Supported resumable learning with SQLite-backed accounts and progress storage, password hashing, and user-scoped session access.

Custom Vulkan Renderer | C++, Vulkan, Slang, GLFW, CMake - Waco, TX **May 2026-Present**

- Architecting a custom **C++ and Vulkan** renderer to deepen experience with real-time graphics, performance, and low-level GPU programming.
- Orchestrated a dynamic-rendering pipeline with swapchain management, GPU vertex and index buffers, descriptor sets, uniform-buffer transforms, and frame synchronization.
- Automated Slang shader compilation to SPIR-V with CMake and added window-resize handling and mouse-driven model rotation.
- Integrating Vulkan-Hpp RAII wrappers and explicit GPU resource management while continuing to extend the renderer.

Personal Projects (continued)

VimAnywhere | Native macOS Utility **Sep 2026**
Swift, SwiftUI, AppKit, CoreGraphics, macOS Accessibility API
github.com/Lukovsky1/VimAnywhereMacOS27

- Developed a native macOS prototype that brings Vim-style navigation and editing to text fields across applications by translating global keyboard input into native editing actions.
- Built a CoreGraphics event-tap pipeline to intercept keyboard events and synthesize cursor movement, text selection, deletion, and undo/redo while preserving macOS Command shortcuts.
- Implemented Normal and Insert modes with a pending-command state machine for multi-key operations such as `dd`, `dw`, `cw`, and `gg`.
- Integrated `NSWorkspace` and `AXUIElement` to inspect the active application's focused control and restrict Vim command handling to text fields and text areas.
- Created a floating, click-through SwiftUI mode indicator hosted in an AppKit window, with permission checks and request flows for Accessibility, Input Monitoring, and synthetic input.